

Class 10

David Majeed (PID:A17885958)

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Background

The main repository of high-resolution data on biomolecules is called the Protein Data Bank (PDB)

PDB Stats

What is in the PDB in terms of molecule type and structure determination method

```
#read the file from the rcsb website and then format properly
pdb<- read.csv("Data Export Summary.csv")
x<-pdb$X.ray
tmp<-sub(",", "", x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

```
#Lets make this a function
```

```
rm.comma<- function(x) {  
  tmp<-sub(",", "", x)  
  sum(as.numeric(tmp))  
}
```

```
rm.comma(pdb$EM)
```

```
[1] 33880
```

```
#we can also use a different import function
```

```
library(readr)  
read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# A tibble: 6 x 9
```

`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1 Protein (only)	180758	23111	12813	348	229	84
2 Protein/Oligosacch~	10488	3741	34	8	11	1
3 Protein/NA	9205	6751	287	26	8	0
4 Nucleic acid (only)	3154	250	1578	3	15	3
5 Other	178	27	35	4	0	0
6 Oligosaccharide (o~	11	0	6	0	1	0

```
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
#lets find the percetange found by x-ray
```

```
n.tot <- rm.comma(pdb$Total)
```

```
n.xray<- rm.comma(pdb$X.ray)
```

```
n.em<- rm.comma(pdb$EM)
```

```
(n.xray+n.em)/n.tot
```

```
[1] 0.9386623
```

```
#How many total proteins?
```

```
pdb[1,9]
```

```
[1] "217,375"
```

```
#there are 202556314 sequences known for proteins
```

```
217375/202556314
```

```
[1] 0.001073158
```

Q1. 93.87% of structures in the PDB are solved by X-Ray and Electron Microscopy

Q2. There are 217,375 proteins; this is only 0.11% of known proteins. With most only known through one method

Q3. There are 1,227 structures that are shown when searching for HIV-1 protease

Visualizing PDB Data With Mol-star

Main stand alone website version



Figure 1: The HIV-protease enzyme is a homodimer with two chains

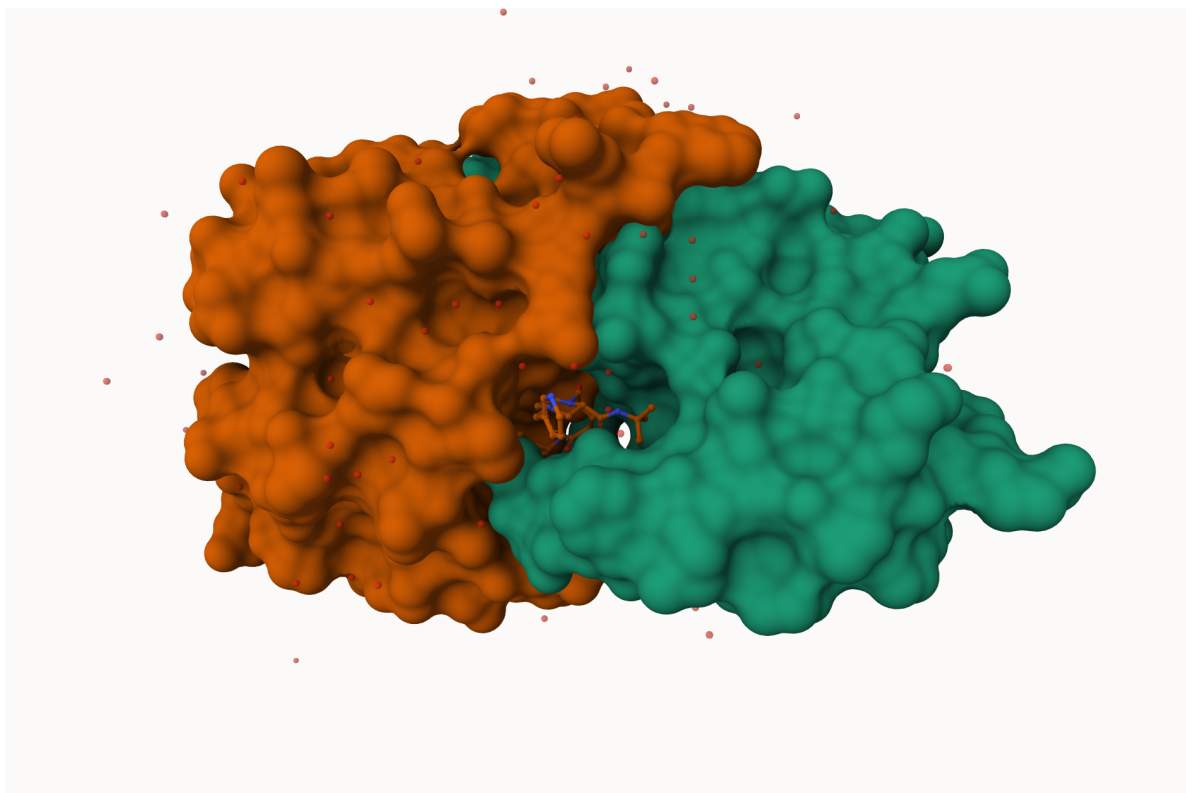


Figure 2: Surface display showing the binding cleft site for the inhibitor (drug) molecule

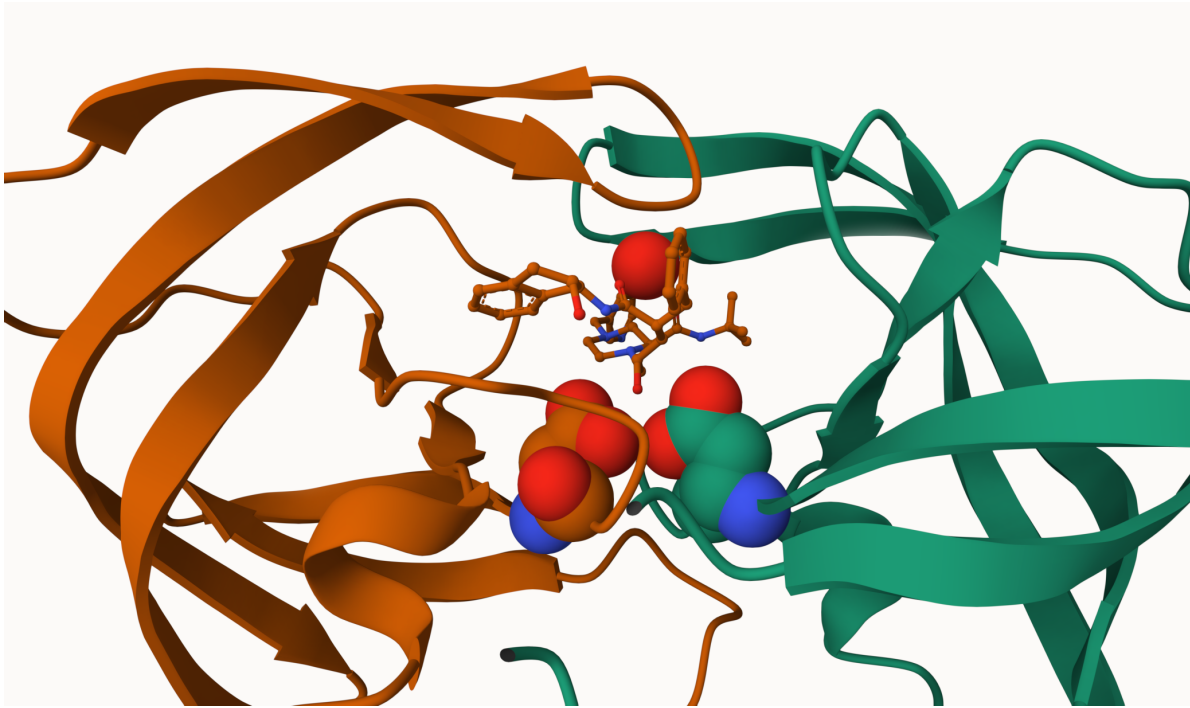


Figure 3: Spacefill display of catalytic ASP25 amino acids and key binding site water molecules

```
#install.packages("pak")
library(bio3d)
#pak::pkg_install(c("bioboot/3dview", "NGLViewR", "bioc::msa"))

pdb<- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
attributes(pdb)
```

```
$names
```

```
[1] "atom"    "xyz"      "seqres"   "helix"    "sheet"    "calpha"   "remark"   "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

There are lots of functions that can work with these `pdb` objects

```
head(pdbseq(pdb))
```

```
  1    2    3    4    5    6  
"P" "Q" "I" "T" "L" "W"
```

```
library(bio3dview)
```

```
#view.pdb(pdb)
```

```
#view.pdb(pdb, colorScheme = "sse", backgroundColor = "black" )
```

```
#Lets highlight the active site ASP, the two chains, and the ligands
```

```
library(NGLViewer)
```

```
#active.site<- atom.select(pdb, resno=25)
```

```
#view.pdb(pdb,
```

```
#       cols = c("red", "blue"),
```

```
#       highlight = active.site,
```

```
#       highlight.style = "spacefill",
```

```
#       backgroundColor = "grey") |>
```

```
#setRock()
```

Let's Predict The Flexibility of A Given Structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a pdb object

```
adk<- read.pdb("6s36")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

```
#summary
```

```
adk
```

Call: read.pdb(file = "6s36")

Total Models#: 1

Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)

Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

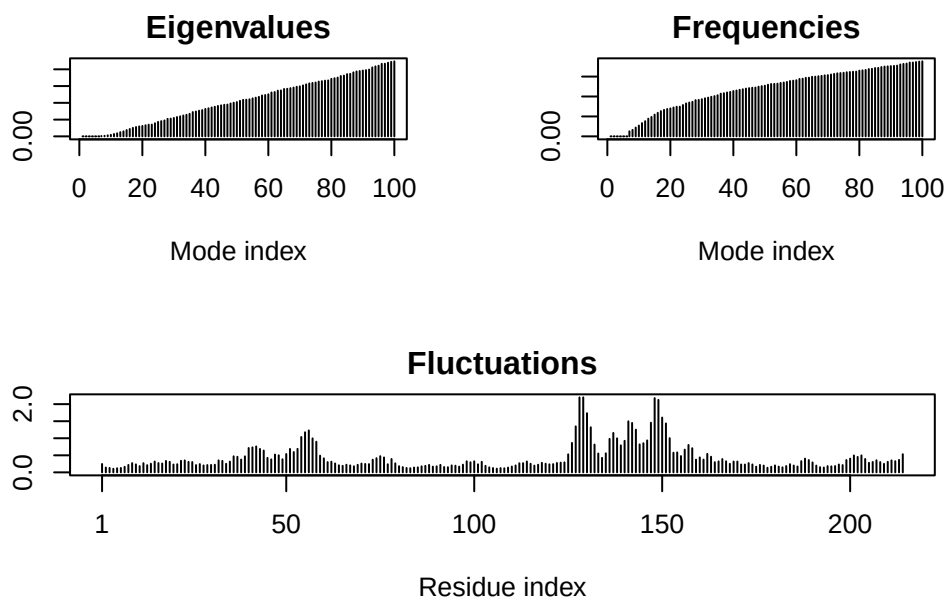
```
m<-nma(adk)
```

Building Hessian... Done in 0.03 seconds.

Diagonalizing Hessian... Done in 0.36 seconds.

```
#make a graph
```

```
plot(m)
```



```
#view.nma(m)
#write out the results of for viewing in Mol-Star
mktrj(m, file="nma.pdb")
```

Comparative Analysis of The ADK Family

Our first step is finding the sequence with the “lake_A” id

```
library(NGLViewer)
library(bio3dview)

#id<-"lake_A"
#aa<- get.seq(id)
#blast<-blast.pdb(aa)
#hits<-plot(blast)
#hits$pdb.id
#files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)

#Align and superpose all these ADK files
#pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
#pdbs
#view.pdbs(pdbs)

#Some code turned into text as R was having issues rendering it as a pdf
```

PCA of all this data coordinates

```
#pc<-pca(pdbs)
#plot(pc, 1:2)

#interactive view
#view.pca(pc)
#mktrj(pc, file="pca.pdb")
```